

TrES-5b

R-0957

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_240630 · created 2026-07-29T15:26:47+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460491.7872 +/- 0.004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.028 +/- 0.054
Transit depth (Rp/Rs)^2	0.076 +/- 0.3 [%]
Orbital Inclination (inc)	80.6 +/- 2.9
Airmass coefficient 1 (a1)	0.0154 +/- 0.0084
Airmass coefficient 2 (a2)	3.0 +/- 0.26
Scatter in the residuals of the lightcurve fit is	41.7 %
Best Comparison Star	None
Optimal Aperture	6.97
Optimal Annulus	8.83
Transit Duration (day)	0.028 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.028 ± 0.054 vs 0.142 (deviation 1.15σ)
Duration fit vs published	0.028 d vs 0.0758 d (deviation 0.97σ)
Mid-time O–C	3.0 min
Depth SNR	0.3
Residual scatter	41.7 %

Light curve

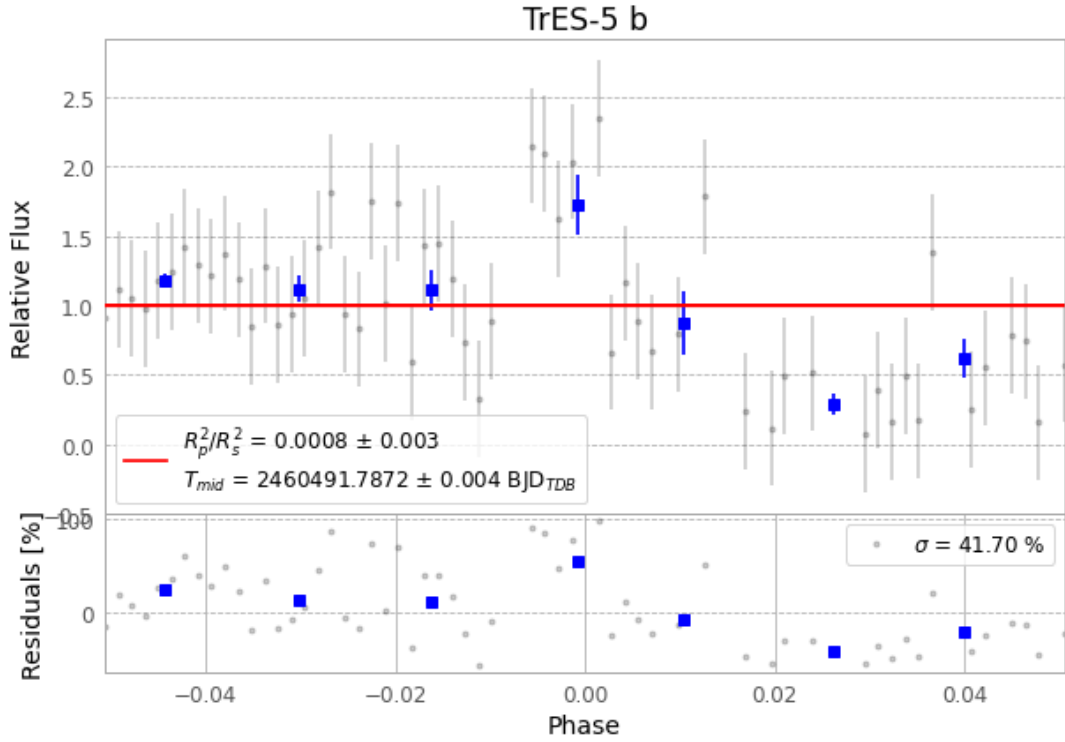


Figure 1 — FinalLightCurve_TrES-5 b_2024-06-29.png (SHA-256 2b3c802342154dcc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [356, 169], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ab398ff586d79b1f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

73 FITS frames (48 MB), TRES-5240630050315.FITS → TRES-5240630083915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-240630.log (ad2070994be0...), FinalLightCurve_TrES-5 b_2024-06-29.png (2b3c80234215...), FinalLightCurve_TrES-5 b_2024-06-29.csv (f5f1c23a2fab...)

Compute resources

Wall time 245.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 15:26Z.